

Reading: Comprehensive Guide to Generative AI

1.4h • Preparing and Strating Evaluation of Generative AI

0h / 1.4h

Introduction

Generative AI (GenAI) has evolved from basic text and image generation to powering complex systems such as AI agents, enterprise automation, and reasoning engines. This guide explores **core concepts, tools, and frameworks** for building modern GenAI applications, including **RAG, multi-agent systems, prompt engineering**, and cutting-edge libraries like LangGraph.

Whether you're developing chatbots, automation workflows, or knowledge systems, this guide provides a roadmap to the latest advancements. It also introduces additional terms not covered in the course videos. These terms are essential for enhancing your understanding of the course concepts.

Core GenAI Concepts & Terminologies

Foundational Concepts

Term	Definition	Examples/Use Cases
LLM	A type of AI model trained on vast amounts of text data to understand and generate human-like language.	GPT-4o, Claude, LLaMA
Prompting	A technique for designing input instructions to guide LLM outputs.	"Write a summary in 3 sentences." "Answer as a cybersecurity expert."
Prompt Templates	Reusable, structured prompts with placeholders for dynamic inputs.	"Explain {concept} like I'm 5 years old."
RAG (Retrieval-Augmented Generation)	Combines retrieval from external knowledge sources with LLM generation to enhance factual accuracy.	Answering questions with real-time data (for example, Aha! Blog). IT
Retriever	A system component designed to fetch relevant information from a database or dataset.	Vector similarity search using FAISS, Elasticsearch
Agent	An autonomous AI system that can plan, reason, and execute tasks using tools.	AutoGPT, LangChain Agents
Multi-Agent System	A framework in which multiple AI agents collaborate to solve complex tasks.	Microsoft AutoGen, CrewAI
Chain-of-Thought	A prompting technique that encourages models to decompose problems into intermediate steps.	"Let's think step by step..."
Hallucination Mitigation	Strategies to reduce incorrect or fabricated outputs from LLMs.	RAG, fine-tuning, prompt constraints
Vector Database	A database optimized for storing and querying vector embeddings.	Pinecone, Chroma, Weaviate
Orchestration	Tools to manage and coordinate workflows involving multiple AI components.	LangChain, Llamaindex
Fine-tuning	Adapting pre-trained models for specific tasks using domain-specific data.	LoRA (Low-Rank Adaptation), QLoRA (quantized fine-tuning)

Tools & Frameworks

Model Development & Deployment

Tool/Framework	Definition	Examples/Use Cases	Reference Link
Hugging Face	A platform hosting pre-trained models and datasets for NLP tasks.	Accessing GPT-2, BERT, Stable Diffusion	Hugging Face IT
LangChain	A framework for building applications with LLMs, agents, and tools.	Creating chatbots with memory and web search	LangChain IT
AutoGen	A library for creating multi-agent conversational systems.	Simulating debates between AI agents	AutoGen IT
CrewAI	A framework for assembling collaborative AI agents with role-based tasks.	Task automation with specialized agents	CrewAI IT
Beakl	A lightweight framework to build production-ready multi-agent systems.	Distributed problem-solving systems	Beakl IT
Llamaindex	A tool to connect LLMs to structured or unstructured data sources.	Building Q&A systems over private documents	Llamaindex IT
LangGraph	A library for building stateful, multi-actor applications with LLMs.	Cyclic workflows, agent simulations	LangGraph IT

Retrieval & Infrastructure

Tool/Framework	Definition	Examples/Use Cases	Reference Link
FAISS	A library for efficient similarity search of dense vectors.	Retrieving top-k documents for RAG	FAISS IT
Pinecone	A managed cloud service for vector database operations.	Storing embeddings for real-time retrieval	Pinecone IT
Haystack	An end-to-end framework for building RAG pipelines.	Deploying enterprise search systems	Haystack IT

Advanced Prompting Techniques

Concept	Definition	Example
Zero-Shot Prompting	Providing examples in the prompt to guide the model's output format.	"Translate to French: 'Hello!' → 'Bonjour!'." "Greetbye!" → "..."
Zero-Shot Prompting	Directly asking the model to perform a task without examples.	"Classify this tweet as positive, neutral, or negative: [tweet]"
Chain-of-Thought	Encouraging step-by-step reasoning.	"First, calculate X. Then, compare it to Y. Find answer: ..."
Prompt Chaining	Breaking complex tasks into smaller prompts executed sequentially.	Prompt 1: Extract keywords — Prompt 2: Generate summary from keywords.

Key Architectures & Workflows

RAG Pipeline

- Retrieval:** Query vector database (for example, Pinecone) for context.
- Augmentation:** Combine context with user prompt.
- Generation:** LLM (for example, GPT-4o) produces final output.

Multi-Agent System

- Agents:** Specialized roles (for example, researcher, writer, critic).
- Orchestration:** LangGraph for cyclic workflows, AutoGen for conversations, and so on.
- Tools:** Web search, code execution, API integrations, and so on.

References

- [Retrieval-Augmented Generation \(RAG\) Page](#) [IT](#)
- [Chain-of-Thought Prompting](#) [IT](#)
- [LangChain Documentation](#) [IT](#)
- [CrewAI Documentation](#) [IT](#)
- [Beakl Key Features](#) [IT](#)

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